Rashedul Alam Adnan

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WORK EXPERIENCE

Research Assistant

Analytical and Environmental Chemistry Laboratory, Binghamton University | Jan 2022 - Present

- Develop and characterize nanomaterial-based biosensors, including polystyrene-coated sensors and SERS-based pH nanoprobes for aerosol measurements.
- Investigate microplastic and nanoplastic detection in aerosols and apply machine learning to spectroscopic data analysis.

Teaching Assistant

Department of Chemistry, Binghamton University | Jan 2022 - Present

- Teach undergraduate general, analytical, and instrumental chemistry laboratories for CHEM 106, 111, 221, and 422.
- Deliver pre-laboratory lectures, supervise experiments, lead discussions, hold office hours, proctor examinations, and grade coursework.
- Guide students in titration, thermochemistry, ICP-OES, HPLC, FAAS, UV-Vis, FTIR, and GC-MS techniques.

Process Engineer

Eastern Refinery Limited | Jul 2017 - Dec 2022

- Operated and optimized atmospheric distillation, catalytic reforming, natural gas condensate processing, fuel desulfurization, and asphaltic bitumen production units.
- Performed quality control of fuel oils and bituminous products to support refinery operations.

Research Assistant

Department of Civil and Environmental Engineering, The Hong Kong Polytechnic University | Dec 2015 - Oct 2016

- Measured air quality and analyzed volatile organic compounds using GC-MS.
- Characterized pollution sources, developed emission inventories, and modeled air pollutant dispersion using AERMOD and HYSPLIT.

Assistant Engineer Chemical

Bashundhara Paper Mills Limited | Nov 2014 - Aug 2015

- Supported wastewater treatment and recycling, deinking plant operations, and effluent treatment plant operations.
- Assessed power plant feedwater quality to support utility operations.

RESEARCH PROJECTS

SERS Based pH Nanoprobes for Aerosol Droplets

- Develop pH-sensitive nanoprobes for aerosol droplet measurements using surface-enhanced Raman spectroscopy and spectroscopic data analysis.

Indoor and Outdoor Air Quality in Hong Kong and Dhaka

- Investigated air pollution using real-time monitoring instruments and GC-MS, including indoor air quality measurements.

Hydrogen Production Plant Simulation

- Simulated a hydrogen production plant in Aspen HYSYS, with a reported simulated yield of 96%.

Sulfuric Acid Plant Process Design

- Designed a 100 metric ton/day sulfuric acid production process and developed a MATLAB algorithm relating reactor height to fractional conversion.

EDUCATION

PhD Candidate in Chemistry

Binghamton University, State University of New York | Jan 2022 - Present

Bachelor of Science in Chemical Engineering

Bangladesh University of Engineering and Technology

TECHNICAL SKILLS

Spectroscopy and characterization: SERS, fluorescence spectroscopy, FTIR, UV-Vis spectroscopy, dynamic light scattering, and cyclic voltammetry.

Analytical instrumentation: GC-MS, HPLC, ICP-OES, and flame atomic absorption spectroscopy.

Programming and data analysis: Python, MATLAB, SPSS, OriginPro, and machine learning for spectroscopic analysis.

Process design and modeling: Aspen HYSYS, AutoCAD, AutoCAD P&ID, Microsoft Visio, AERMOD, and HYSPLIT.

PUBLICATIONS AND PRESENTATIONS

Amin, R.; Adnan, R. A.; Islam, M. R. “Synthesis and Optimization of PVA Based Paper Lamination Adhesives.” International Journal of Innovation and Applied Studies, 2014, 8(4), 1836-1841.

Adnan, R. A.; Rahman, M.; Hung, W. T.; Lee, S. C. “Investigation of Indoor Air Pollution in Urban and Rural Kitchens of Bangladesh.” ICChe conference proceedings, Dhaka, Dec 2017.

Adnan, R. A. “pH Detection of Aerosol Droplets Using SERS Based pH Nanoprobes.” Poster presentation, SNO Conference, Rhode Island, Nov 9, 2024.

AWARDS AND LEADERSHIP

- Student Poster Award, SNO Conference, Rhode Island, Nov 2024.
- International Visitor Leadership Program award in air quality, U.S. Department of State, Washington, DC, Jul 2018.
- Dhaka Education Board Scholarship, Higher Secondary School Certificate Examination, 2006.
- President, Bangladesh Graduate Student Association, Aug 2023 - 2024; Summer Chemistry Workshop Facilitator, 2022 and 2023.

PROFESSIONAL AFFILIATIONS

Lifetime Member, Institute of Engineers Bangladesh, M40455; Member, Chemical Engineering Association, BUET, 2009-2014; Research Fellow, Fredskorpset, Norway, 2015-2017; Member, Clean Air Asia Project, 2015-2017.